

Thermodynamics

Lecture 2C:

- Energy Transfer, General Energy Analysis
- Pure substances

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After this lecture

- Understand energy analysis of open systems and cycles.
- Define energy conversion efficiency.
- Introduction to pure substances and phase change processes.

Learning objectives

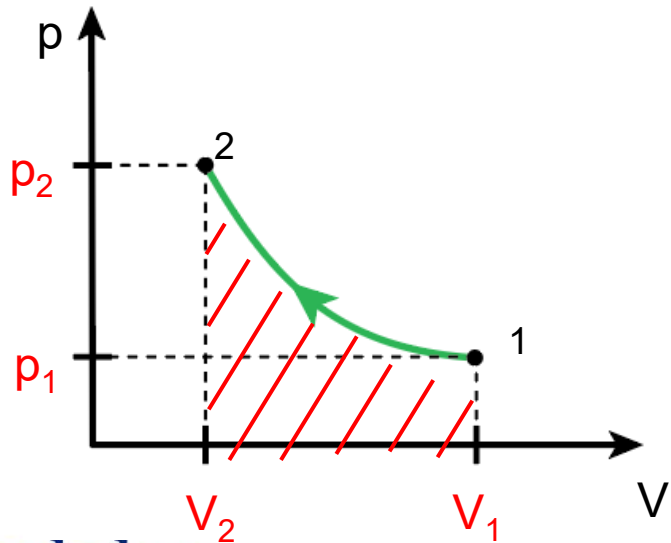
What you will be able to do after this course:

- I. *Knowledge:* Write the first and second laws of thermodynamics.**
- II. *Comprehension:* Describe the first and second laws in the student's own language.**
- III. *Application:* Solve simple single-answer problems using the first law.**
- IV. *Analysis:* Solve problems requiring both the first and second laws.*
- V. *Synthesis:* Develop schemes to improve the efficiency or performance of thermodynamic systems, e.g., compressors, turbines, heat pumps, etc.*

Recap Lecture 2B

Boundary work

$$W = \int_1^2 p dV$$



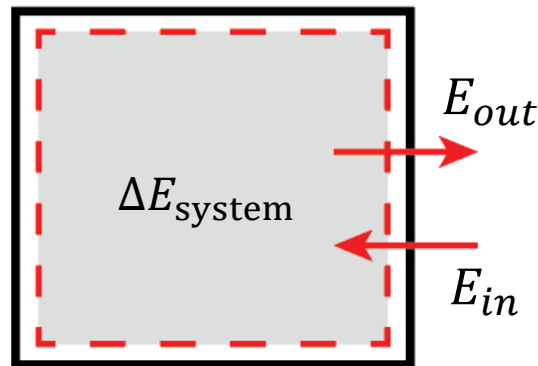
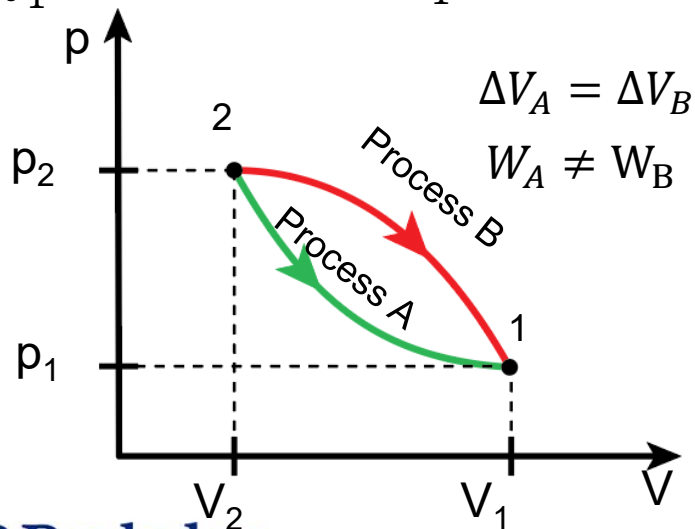
Inexact differentials $\delta Q; \delta W$

Exact differentials dV

Recap Lecture 2B

W and Q are path dependent

$$\int_1^2 dV = V_2 - V_1 \quad \int_1^2 \delta W = W_{12}$$



$$E_{in} - E_{out} = \Delta E_{system}$$

$$\Delta U = Q - W$$

Energy transfer in/out of a closed system

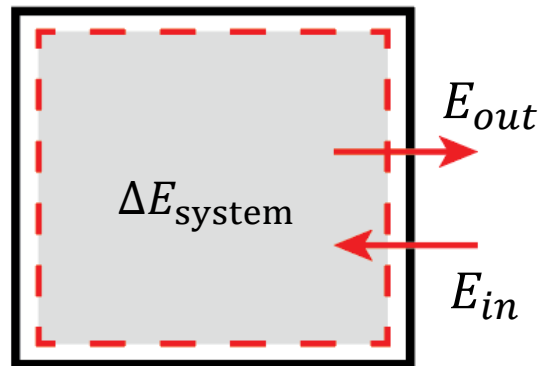
- Account for energy transfer with surroundings.

$$E_{in} - E_{out} = (Q_{in} - Q_{out}) + (W_{in} - W_{out})$$

Q $-W$

Heat in: $Q > 0$

Work out: $W > 0$



1st Law of thermodynamics: Example II

- Determine U_2 of the air inside the box.

- Define the system

$$\Delta U = Q - W$$

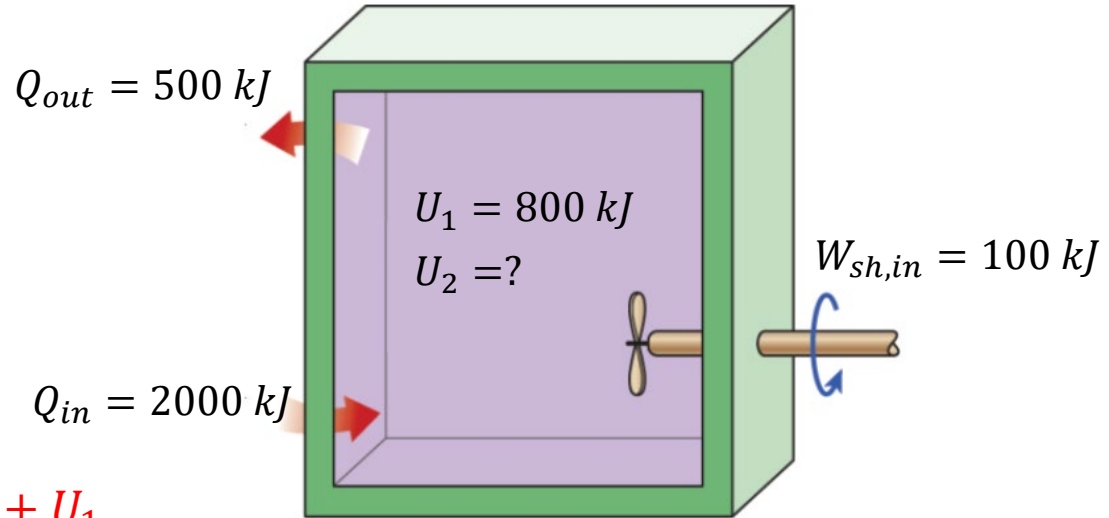
$$W = -100 \text{ kJ} ;$$

$$\Delta U = U_2 - U_1$$

$$Q = Q_{in} - Q_{out} = 1500 \text{ kJ} \quad Q_{in} = 2000 \text{ kJ}$$

$$U_2 = Q_{in} - Q_{out} + W_{shaft,in} + U_1$$

$$U_2 = Q_{in} - Q_{out} + W_{shaft,in} + U_1 = 2400 \text{ kJ}$$



Recap exercise: Isochoric process

Question:

1. What is the work extracted during a process of isochoric heating in a cylinder from state 1 to state 2:

To Do:

- Determine the relevant equation for Work.
- Simplify and rewrite that equation
- Go to pollev.com/claudiohail
- ... and answer question 1.



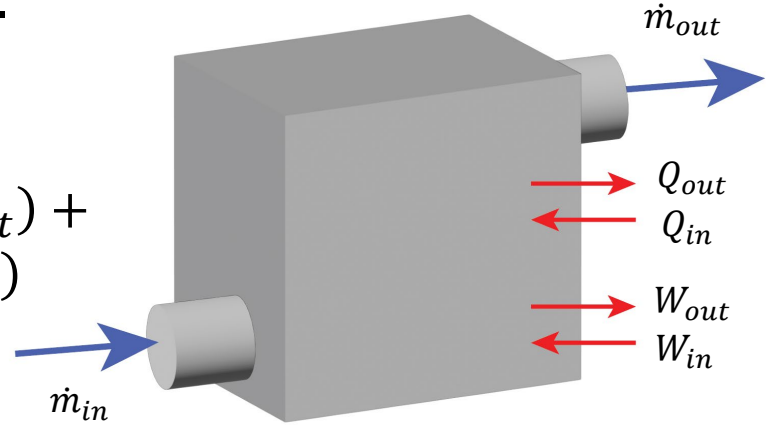
1st Law of thermodynamics: Open system

- Mass flows carry energy in/out.

$$E_{in} - E_{out} = \overset{Q}{(Q_{in} - Q_{out})} + \overset{-W}{(W_{in} - W_{out})} + \dots (E_{mass,in} - E_{mass,out})$$

$$\Delta E_{\text{system}} =$$

$$\text{Rate formulation: } \dot{E}_{in} - \dot{E}_{out} = \frac{dE_{\text{system}}}{dt} = 0$$



Energy analysis of thermodynamic cycles

- A closed system undergoing a thermodynamic cycle.

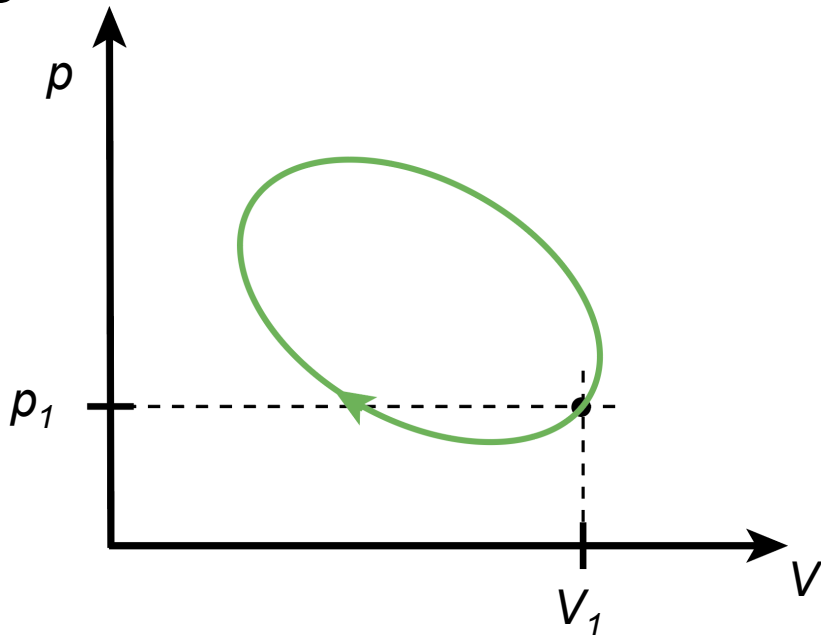
Initial state = final state

$$\Delta E_{System} = E_2 - E_2 = 0$$

Therefore

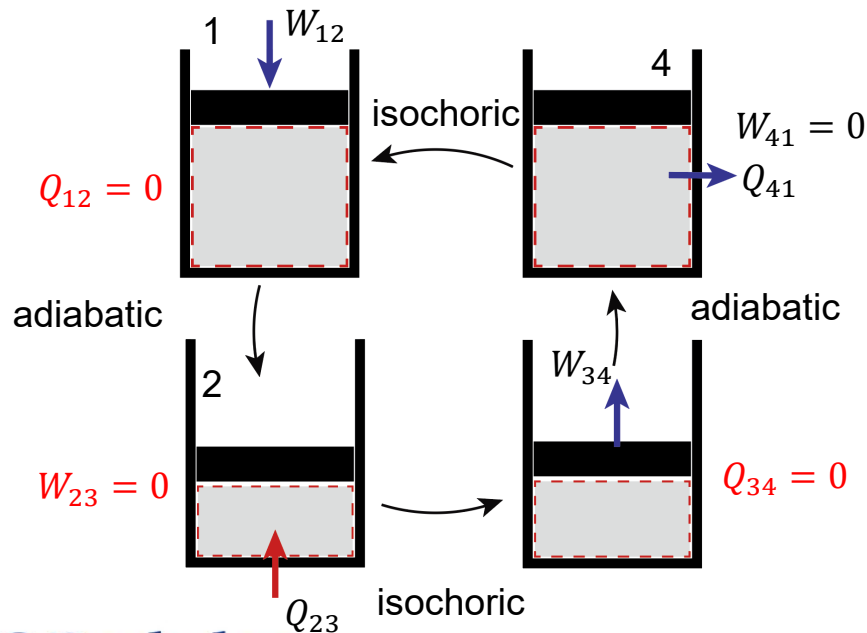
$$E_{in} - E_{out} = 0$$

$$W_{net,out} = Q_{net,in}$$



Energy analysis of thermodynamic cycles

- Example of a closed cycle process of gas in a piston



$$W_{net,out} = Q_{net,in}$$

$$W_{34} - W_{12} = Q_{23} - Q_{41}$$

Energy conversion efficiencies (I)

- Denoted by Greek letter η

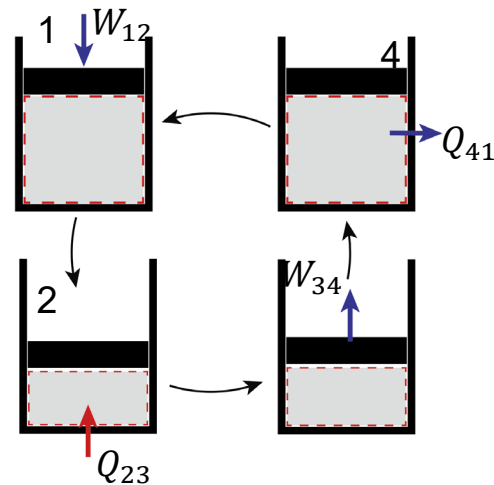
$$\text{Efficiency} = \frac{\text{Desired output}}{\text{Required input}} \leq 1$$



Heater:

$$\eta = \frac{Q_{out}}{W_{in}}$$

$$\eta = \frac{Q_{out}}{Q_{in}}$$

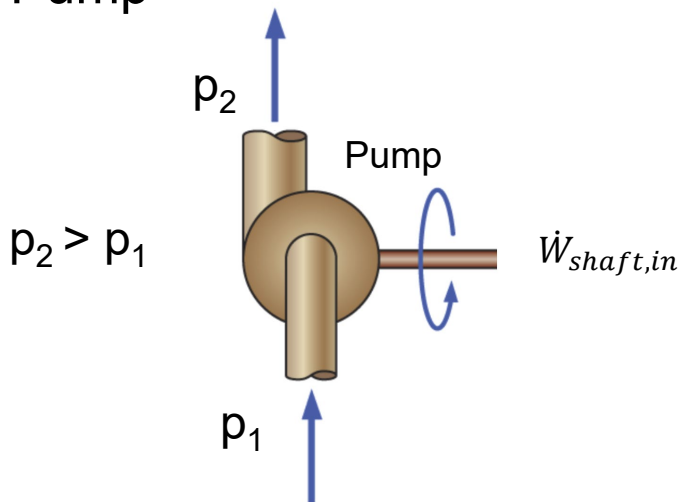


Heat engine:

$$\eta = \frac{W_{net}}{Q_{in}} = \frac{Q_{in} - Q_{out}}{Q_{in}} = 1 - \frac{Q_{out}}{Q_{in}}$$

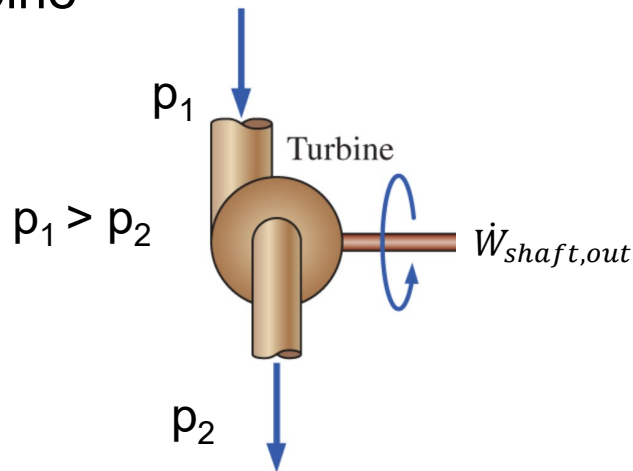
Energy conversion efficiencies (II)

- Pump



$$\eta_{pump} = \frac{\text{Mech. energy increase fluid}}{\text{Mech. energy input}} = \frac{\Delta \dot{E}_{mech,fluid}}{\dot{W}_{shaft}}$$

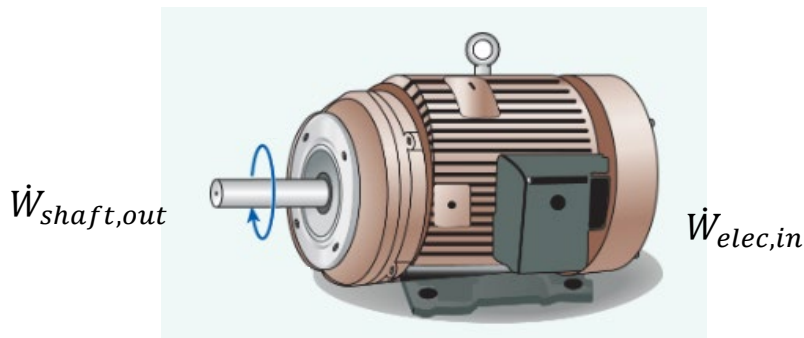
- Turbine



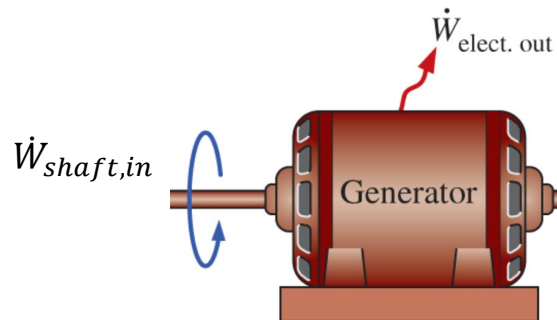
$$\eta_{turbine} = \frac{\text{Mech. energy output}}{\text{Mech. energy decrease fluid}} = \frac{\dot{W}_{shaft,out}}{\Delta \dot{E}_{mech,fluid}}$$

Energy conversion efficiencies (III)

- Motor



- Generator



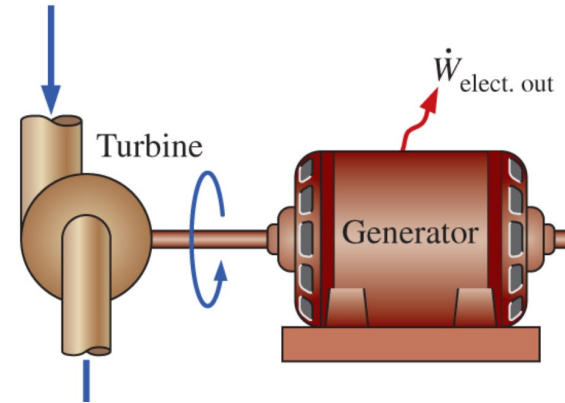
$$\eta_{motor} = \frac{\text{Mechanical power output}}{\text{Electric power input}} = \frac{\dot{W}_{shaft,out}}{\dot{W}_{elec,in}}$$

$$\eta_{generator} = \frac{\text{Electric power output}}{\text{Mechanical power input}} = \frac{\dot{W}_{elec,out}}{\dot{W}_{shaft,in}}$$

Energy conversion efficiencies

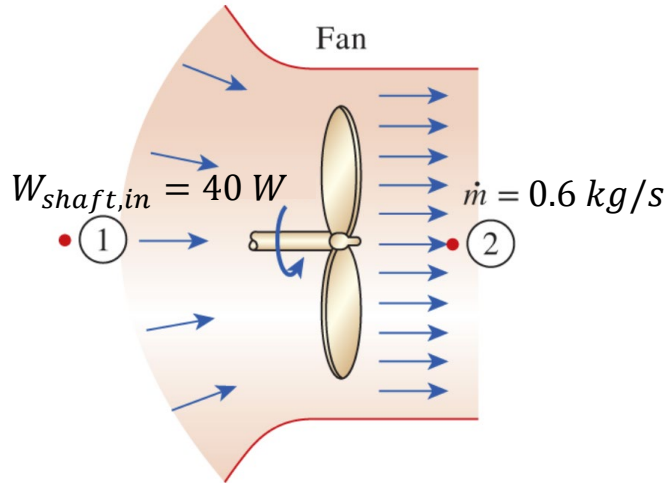
- Combining efficiencies of multiple devices in series:

$$\eta_{tot} = \frac{\dot{W}_{shaft,out}}{\Delta \dot{E}_{mech,fluid}} \frac{\dot{W}_{elec,out}}{\dot{W}_{shaft,in}} = \eta_{turbine} \eta_{generator}$$



Group Work: in groups of 2; solve the following problem.

Given:



$$V_1 = 0 ; V_2 = 10 \text{ m/s}$$

$$z_1 = z_2$$

$$P_1 \approx P_{atm} ; P_2 \approx P_{atm}$$

Question:

1. What is the efficiency of this device?
Give a formula and value.

To Do:

- Define the thermodynamic system of interest.
- Determine the change in energy of the fluid.
- Determine the efficiency.
- Go to pollev.com/claudiohail
- ... and answer question 1.

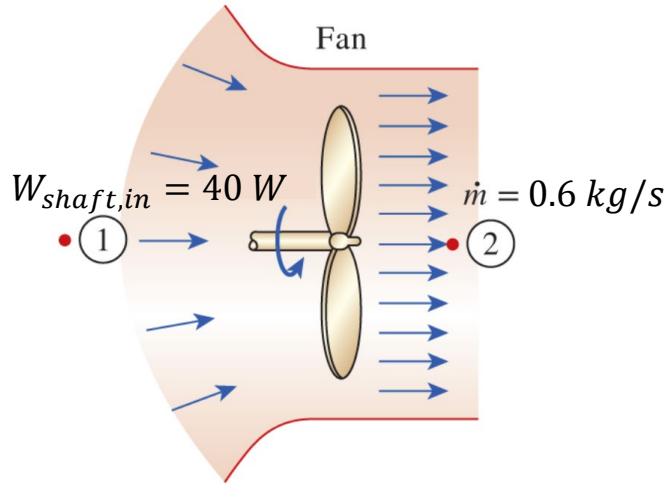
$$\eta = \frac{\Delta \dot{E}_{mech,fluid}}{\dot{W}_{shaft,in}} = \frac{\dot{m} V_2^2 / 2}{\dot{W}_{shaft,in}} = 0.75$$

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Group Work: in groups of 2; solve the following problem.

Given:



$$V_1 = 0 ; V_2 = 10 \text{ m/s}$$

$$z_1 = z_2$$

$$P_1 \approx P_{atm} ; P_2 \approx P_{atm}$$

$$\eta = \frac{\Delta \dot{E}_{mech,fluid}}{\dot{W}_{shaft,in}} = \frac{\dot{m} V_2^2 / 2}{\dot{W}_{shaft,in}} = 0.75$$

$$\Delta \dot{E}_{mech,fluid} = \Delta U + \Delta KE + \Delta PE = \frac{1}{2} \dot{m} (V_2^2 - V_1^2)$$

$$\eta = \frac{\dot{m} V_2^2 / 2}{\dot{W}_{shaft,in}} = 0.75$$

Summary: Energy & General Energy Analysis

Mass flow:

$$\dot{m} = \rho \dot{V} = \rho A_c V_{avg}$$

Mechanical energy:

$$\dot{E}_{mech} = \dot{m} e_{mech} = \dot{m} \left(\frac{P}{\rho} + \frac{V^2}{2} + gz \right)$$

Electrical work

$$W_e = UI \Delta t$$

Shaft work

$$W_{sh} = 2\pi n T$$

Spring work

$$W_{spring} = \frac{1}{2} k (x_2^2 - x_1^2)$$

Energy balance:

$$E_{in} - E_{out} = \Delta E_{system}$$

Energy balance (rate):

$$\dot{E}_{in} - \dot{E}_{out} = \frac{dE_{system}}{dt}$$

1st law closed system:

$$\Delta U = W - Q$$

Energy conversion efficiency:

$$Efficiency = \frac{\text{Desired output}}{\text{Required input}} \quad \eta = \frac{W_{net}}{Q_{in}}$$

Summary

- Understand energy analysis of open systems and cycles.
- Define energy conversion efficiency.

Next up

- Introduced pure substances and respective phase change processes.
- Thermodynamic property diagrams for pure substances.
- Procedures on determining the thermodynamic properties of pure substances.